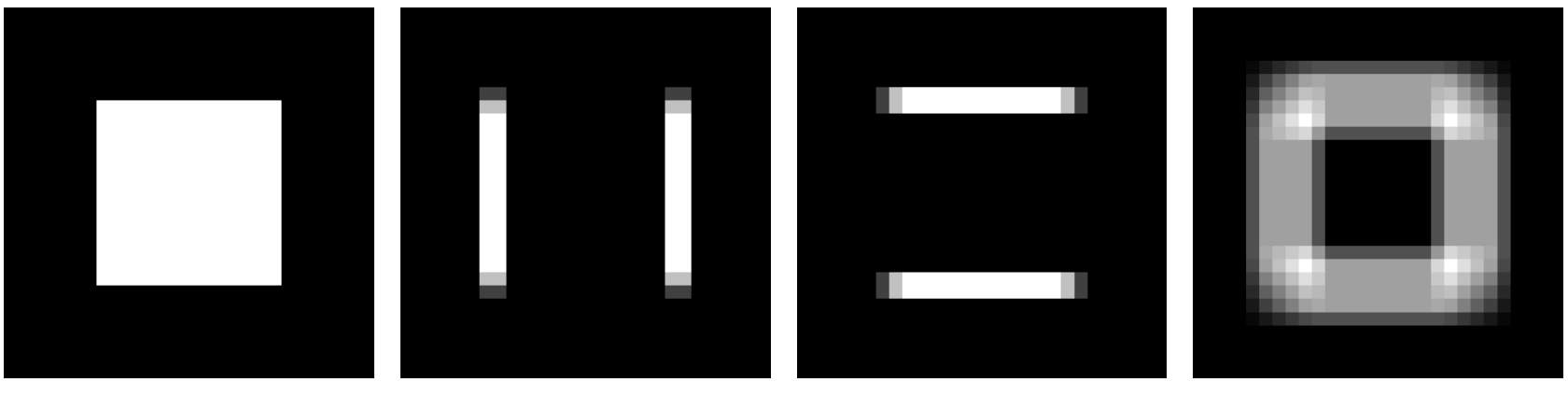
input
vertical-edge expert
horizontal-edge expert
their reader: corners